

Akshay Dahiya

Protocol Engineer with 8+ years building blockchain infrastructure, smart contracts, and distributed systems.

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TECHNICAL SKILLS

Blockchain & Protocol	Solidity, EVM Architecture, Consensus Systems, EIP-712 Relayers, Tokenomics, DeFi, IPFS
Languages & Backend	Go, Python, Node.js, Rust, PostgreSQL, Redis, Docker, AWS, Linux
Web3 Tools	Hardhat, Foundry, Ethers.js, Web3.py
AI & Machine Learning	PyTorch, TensorFlow, Reinforcement Learning (PPO), Computer Vision, Agentic AI
Frontend	React, Next.js, TypeScript, JavaScript

KEY HIGHLIGHTS

- **800 ETH raised** and **5,300+ SBT mints** via protocol staking contracts
- **1M+ gasless transactions** sponsored through custom EIP-712 relayers
- **50+ audited smart contracts** deployed across major EVM networks
- **3M+ daily requests** served by high-scale API gateway infrastructure
- **2 ACM publications**; AI research featured in Forbes
- **2,000+ projects reviewed**, mentoring 500+ aspiring engineers

PROFESSIONAL EXPERIENCE

Lead Protocol Engineer

POWERLOOM PROTOCOL

Dec 2022 - Present

Remote

- Architected protocol migration to DSV network, designing consensus logic for trustless verification
- Engineering 'Snapshotter' node registry contracts, securing **5,300+ nodes** and **800 ETH TVL**
- Designed **\$POWER tokenomics**, staking modules, and airdrop mechanics on Arbitrum Nitro
- Built high-throughput **EIP-712 relayers** ensuring gasless txs for **1M+ interactions**
- Developed **Onchain Points** and Generative Prediction Markets using live protocol data
- Led engineering for **3 incentivized testnets** with CoinList, Polygon zkEVM, and Socket
- Launched **BDS (Block Data Service)**: decentralized real-time Uniswap data market with IPFS proofs

Senior Software Engineer

HYPERSONIX INC.

Mar 2020 - Dec 2022

Remote

- Engineered high-availability **API Gateway** scaling to **3M+ daily requests** for enterprise data
- Optimizing API latency by **80%** using Brotli compression and query tuning
- Implemented enterprise auth with Auth0 (SSO, OAuth 2.0) to secure sensitive endpoints
- Deployed comprehensive ELK observability stack for real-time monitoring

Blockchain Developer

FREELANCE

Dec 2020 - Nov 2022

Remote

- Delivered **50+ audited smart contracts** across EVM chains, prioritizing gas optimization
- Architected complex tokens (ERC-20/721/1155) with deflationary and dynamic reward logic
- Built DeFi protocols including MEV-resistant vaults and automated referral systems
- Engineered Play-to-Earn ecosystems integrated with Chainlink VRF and cross-chain bridges

Graduate Programmer

FIDELITY INTERNATIONAL

Jun 2018 - Nov 2018

Bangalore, India

- Built real-time survey platform using WebSockets, Django, Redis, and D3.js
- Developed React-based analytical tools for behavioral finance portfolio managers

Technical Mentor & Project Reviewer

Feb 2017 - Mar 2020

UDACITY

Remote

- Reviewed **2,000+ technical projects** across Full Stack and AI nanodegrees
- Mentored **500+ students** globally through code reviews and debugging sessions

Open Source Contributor & Admin

Feb 2018 - Aug 2021

GOOGLE SUMMER OF CODE — HYDRA ECOSYSTEM

Remote

- Core contributor to Hydra Ecosystem — framework for smart, self-describing Web APIs
- Served as GSoC Admin/Mentor (2018–2021), guiding **10+ contributors** on open-source tooling

EDUCATION

Master of Science in Computer Science — Machine Learning Specialization

2019 - 2021

GEORGIA INSTITUTE OF TECHNOLOGY

Atlanta, GA

- Research: Built **Examinator** (ML plagiarism detection) and **JackMarker**; published 2 ACM papers
- Coursework: Machine Learning, Reinforcement Learning, Computer Vision, AI, Graduate Algorithms

Bachelor of Technology

2014 - 2018

DCRUST

India

- Comprehensive training in electronics, communication systems, and core engineering principles
- Strong foundations in signal processing, embedded systems, and technical problem-solving

Nanodegree Certifications

2016 - 2020

UDACITY

Online

- **Deep Learning (2020):** Neural networks with PyTorch — CNNs, RNNs, LSTMs, GANs
- **Computer Vision (2020, NVIDIA):** YOLO, R-CNN, Visual SLAM, Image Captioning
- **React (2017):** 3 full-stack apps with Redux — MyReads, Would You Rather, Mobile Flashcards
- **Full Stack (2016):** PostgreSQL, RESTful APIs, OAuth, AWS deployment

Open Courseware Foundations

2015 - 2019

MIT OCW / STANFORD / BERKELEY / EDX

Online

- **MIT 6.006:** Core algorithms — sorting, heaps, BSTs, graph algorithms, dynamic programming
- **MIT 6.042J:** Discrete math — proofs, graph theory, probability
- **Stanford CS161:** Divide-and-conquer, randomized algorithms, NP-completeness
- **Berkeley CS188:** AI fundamentals — search, MDPs, reinforcement learning, Q-learning

PUBLICATIONS

1. Dahiya, A., Graziano, R., Irish, I., Starner, T.

JackMarker with GitDown: A Framework to Counter Plagiarism at Scale

ACM Learning @ Scale 2021. DOI: 10.1145/3430895.3460982

2. Apoorv, R., Dahiya, A., Graziano, R., Starner, T.

Examinator: A Plagiarism Detection Tool for Take-Home Exams

ACM Learning @ Scale 2020. DOI: 10.1145/3386527.3406723